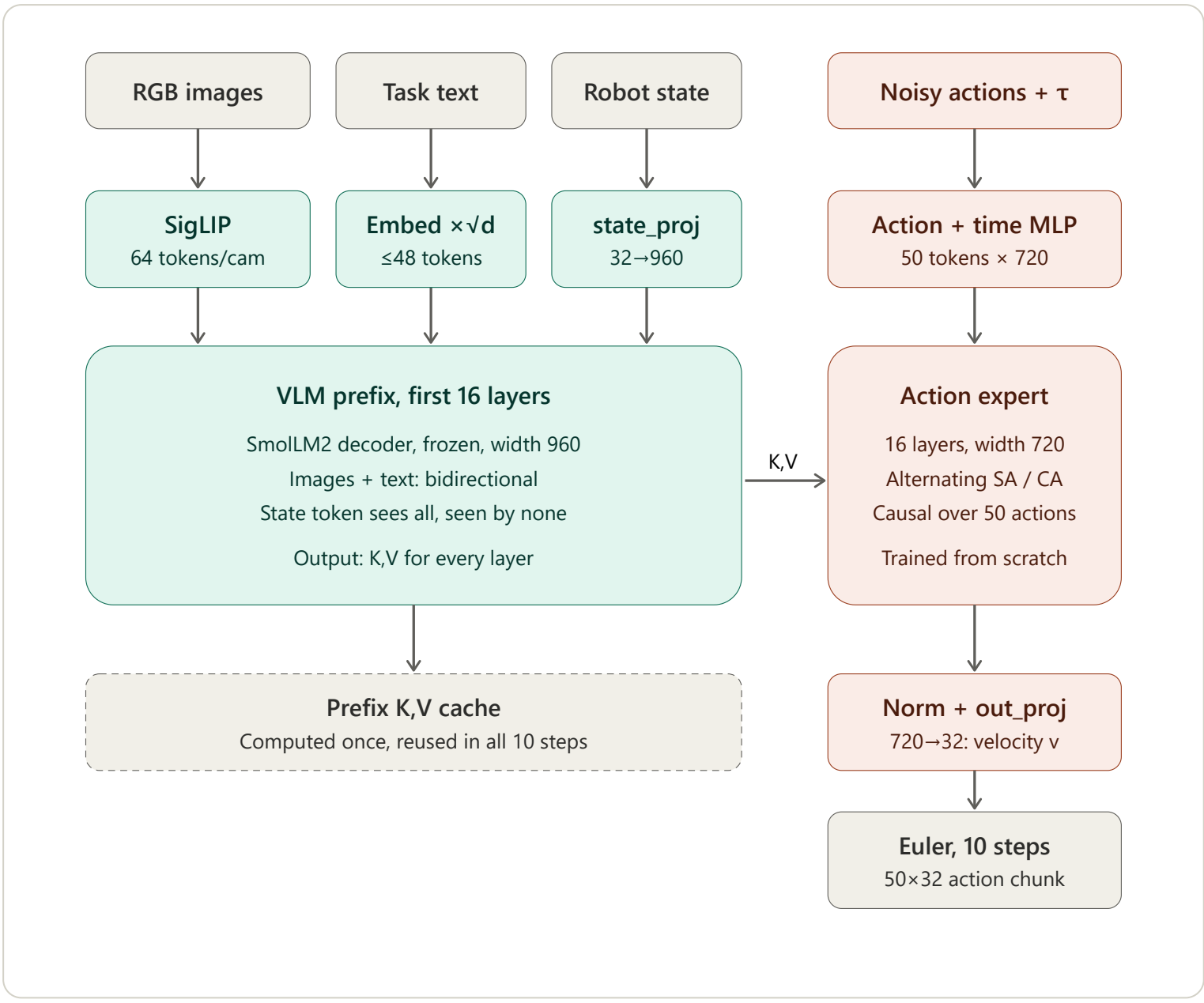
SmolVLA architecture

From the full system down to the inside of a single action-expert layer. Based on the LeRobot implementation.

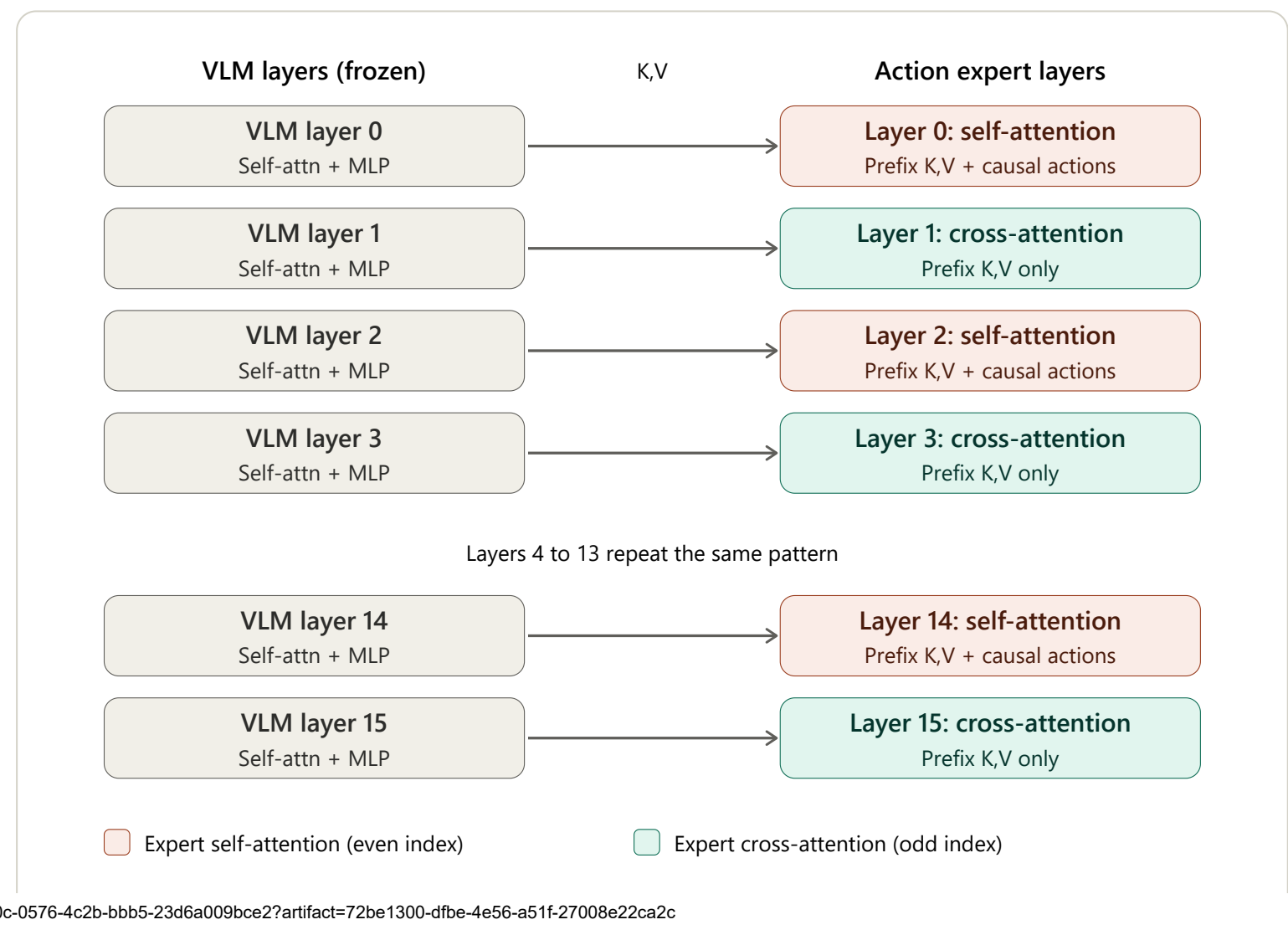
1. System overview

Images, text and state form a prefix run through 16 frozen VLM layers. The action expert denoises 50 actions against the cached prefix K,V, with 10 Euler steps.



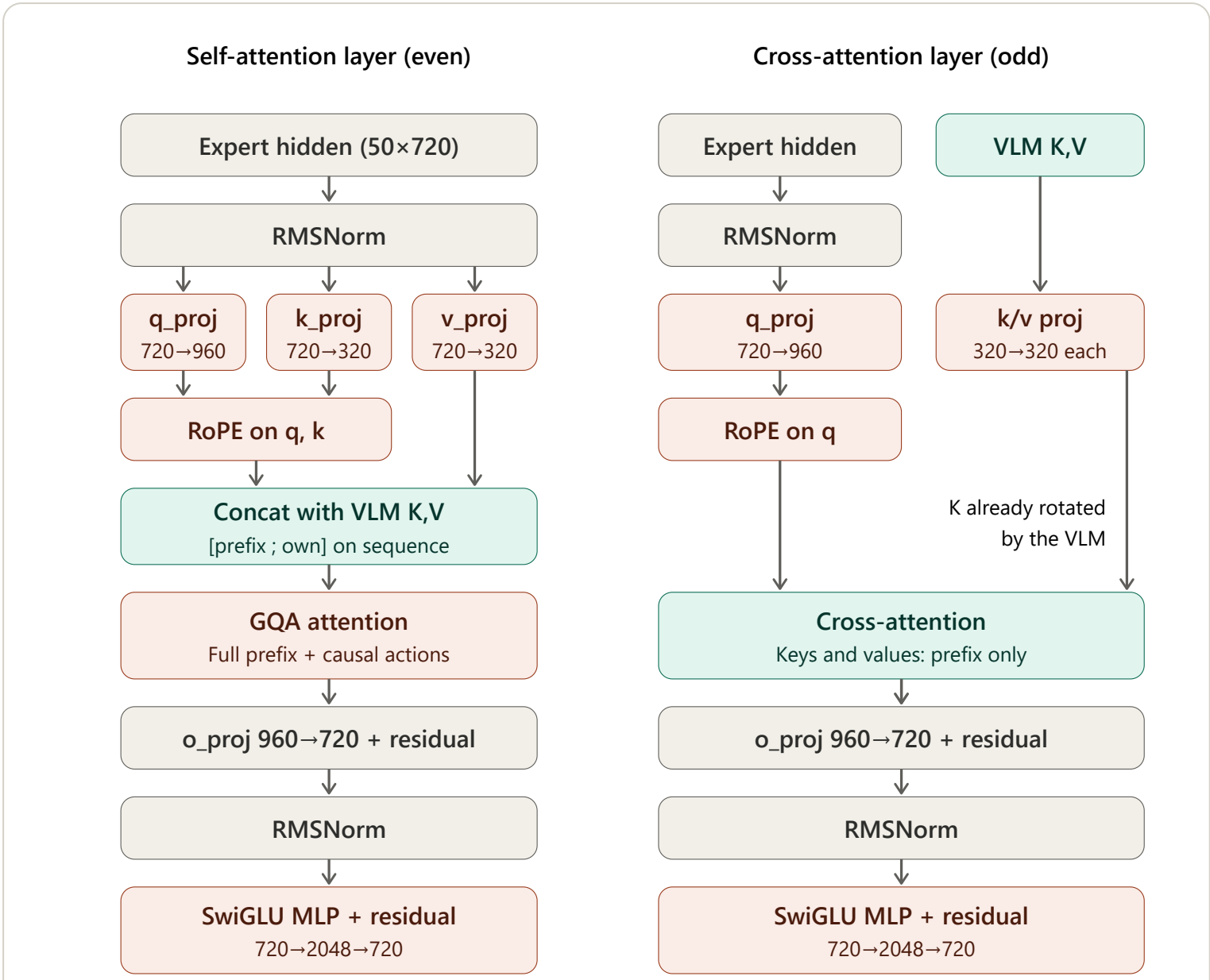
2. Layer stack

Each expert layer reads the K,V of the VLM layer with the same index. Even layers are self-attention, odd layers are cross-attention.



3. Inside one expert layer

Both variants are Llama-style blocks. They differ in where K and V come from and which tokens the queries may see.



Widths 960, 720, 320 and the 2048 MLP size are derived from the SmolVLM2-500M config and an expert width multiplier of 0.75. Verify against your checkpoint.